

Assignment 2: Circular Driving using Reinforcement Learning

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1 Introduction

Our main objective is to design the environment, define suitable rewards, and analyze which RL algorithms are appropriate for solving the task. Proximal Policy Optimization (PPO) is explored as the primary learning algorithm.

2 MDP Formulation

2.1 State Space

The state of the car at any time step is defined as:

$$s = (x, y, v, \theta)$$

where:

- (x, y) is the position of the car
- v is the velocity
- θ is the heading angle

This representation captures positional and other important information necessary for control.

2.2 Action Space

The action space is continuous and consists of:

- Steering angle δ
- Commands like accelerate, decelerate, or constant

Due to continuous actions, policy-based methods are preferred.

2.3 Transition Dynamics

The vehicle motion follows simple kinematic equations:

$$x_{t+1} = x_t + v_t \cos(\theta_t) \Delta t$$

$$y_{t+1} = y_t + v_t \sin(\theta_t) \Delta t$$

$$\theta_{t+1} = \theta_t + \frac{v_t}{L} \tan(\delta) \Delta t$$

3 Reward Function

The reward function is designed to encourage:

- Staying close to the center of the track
- Making progress along the circular path
- Maintaining higher speeds

The reward is defined as:

$$R = -|r - r_{mid}| + \lambda \Delta \phi + \mu v$$

A large negative penalty is applied if the car exits the track.

4 Choice of RL Algorithm

Several RL approaches were considered:

Value-based methods are unsuitable due to continuous action space. Actor-Critic methods provide better stability and efficiency.

5 Model Selection and Justification

The selection of a suitable reinforcement learning model was guided by foundational RL concepts studied from standard resources such as Sutton and Barto, along with introductory articles and lecture videos shared.

Initially, value-based methods such as Q-learning and Deep Q-Networks (DQN) were considered however, these methods require discretization of the action space. Since steering in a driving task is inherently continuous, discretizing steering angles resulted in unnatural and unstable control behavior.

Policy Gradient methods were then studied, as they naturally handle continuous action spaces. While they offer smoother control, vanilla policy gradient methods suffer from high variance during training, which can make learning unstable without careful tuning.

Actor-Critic methods combine the strengths of both value-based and policy-based approaches by learning a policy and a value function simultaneously. Among these methods, Proximal Policy Optimization (PPO) was selected due to its stable learning behavior achieved through constrained policy updates using a clipped objective.

PPO has been widely used in continuous control problems such as robotics and navigation, and its conceptual simplicity made it suitable for this assignment. Therefore, PPO was chosen as a balanced and practical approach for solving the circular driving task.

6 PPO Training Procedure (Pseudocode)

The overall training procedure followed for PPO is summarised below. The pseudocode reflects the logic used in the implementation.

```
Initialize policy network (Actor) and value network (Critic)
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```
for each episode do
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Reset environment
Initialize empty trajectory buffer

while episode not terminated do
    Observe current state
    Sample action from current policy
    Execute action in environment
    Store (state, action, reward, value, log-probability)
end while

Compute discounted returns
Compute advantages using returns and value estimates

for a fixed number of epochs do
    Update policy using clipped PPO objective
    Update value function using mean squared error loss
end for
end for

```

7 Conclusion

This assignment demonstrates how a driving task can be modeled as an MDP and solved using reinforcement learning. By carefully designing the reward function and choosing an appropriate algorithm like PPO, the agent is able to learn stable and efficient driving behavior on a circular track.

References

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